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SUBJECT CODE NO: - RNP-8005
FACULTY OF SCIENCE AND TECHNOLOGY
F. E. (NEP) [Group B]Comp., AI &ML,EEE
Examination January/February 2025
Principles of Programming Language

[Time:3:00 Hours]**[Max. Marks:60]**

- N. B Please check whether you have got the right question paper.
- 1) Use of non-programmable calculator is allowed
 - 2) Q.no. 1 is compulsory.
 - 3) Solve any four questions from Q.no. 2, 3, 4, 5, 6 and 7

Q1 Solve any ten from following questions (any ten)**20**

- A. Which of the following is a high-level programming language?
 - a) Assembly
 - b) Python
 - c) Machine Code
 - d) Binary Short
- B. What does the acronym IDE stand for in programming?
 - a) Integrated Development Environment
 - b) Internal Debugging Engine
 - c) Interactive Development Editor
 - d) Internet Development Extension
- C. What is the primary function, of a compiler?
 - a) To execute a program line by line
 - b) To convert high-level code into machine code
 - c) To debug a program
 - d) To run an interpreted program
- D. Procedural Programming is which type of programming paradigm.
 - a) Imperative
 - b) declarative
 - c) Relational
 - d) Parallel
- E. Which loop is guaranteed to execute at least once?
 - a) for loop
 - b) while loop
 - c) do-while loop
 - d) None of the above
- F. What will be the output of the following C code?


```
#include <stdio.h>

void main()
{
    int x = 5;
    if (x < 1)
        printf("hello");

    if (x == 5)
        printf("hi");
```

else

```
printf("no");
```

```
}
```

a) hi

b) hello

c) no

d) error

G. The C code for (;) represents an infinite loop. It can be terminated by _____

a) break

b) exit(0)

c) abort()

d) terminate

H. What will be the output of the following C code?

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
do
```

```
printf("In while loop ");
```

```
while (0);
```

```
printf("After loop\n");
```

```
}
```

a) In while loop

b) In while loop After loop

c) After loop

d) Infinite loop

I. String constants should be enclosed between _____

a) Single quotes

b) Double quotes

c) Both a and b

d) None of these

J. "Go to" is used for

a) Changing Program

b) Transfer control from one point to another point in a program

c) Entering into new loop

d) Exit from the loop

K. Which of the following correctly declares an array?

a) int anarray[10];

b) int anarray;

c) anarray{10};

d) array anarray[10];

L. What is the output of the following program?

```
void main()
```

```
{
```

```
int a=500,b,c;
```

```
if(a>=400)
```

```
b=300;
```

```
c=200;
```

```
printf("%d %d",b,c);
}
```

- a) 300 200 b) 200 300 c) 400 200 d) 300 400

```
M. void main() {
    int i = 4;
    switch (i)
    {
        default::
        case 3:
            i+= 5;
            if (i=8)
            {
                i++;
                if (i = 9) break;
                i *= 2;
            }
            i=4; break;
        case 8:
            i+= 5;
            break;
    }
    printf("i = %d\n", i);
}
```

What will the output of the sample code above be?

- a) i = 5 b) i=8 c) i=9 d)i=10

- Q2** A) Define Computer? Describe characteristics of computer. **6**
 B) Write a short note ternary operator and relational operator in C **4**
- Q3** A) State and explain different programming domains. **5**
 B) Write a program in C to read principal amount, rate of interest and time duration in years from user calculate simple interest and display it in exponential form. Also, draw flowchart for the same. **5**
- Q4** A) Define 2-dimentional Array. Also write a program to perform addition of two matrix **6**
 B) Differentiate between structure and union. **4**
- Q5** A) List and explain types of conditional statements in C. **5**
 B) Define function. Explain function declaration, function definition with example **5**
- Q6** A) Explain role of \0 (null character) character in string. Write a program in C to accept two strings from user and compare them using built in string library function. **5**
 B) State difference between array and structure. Also, create structure store information of 5 students like student name, Branch, Roll no. and branch for 5 students. **5**

- Q.7**
- | | |
|---|----------|
| A) Write a program to check whether given number is Palindrome or not | 5 |
| B) Write a C program to Swap two number using function (use function call by value) | 5 |